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Credit Default Swaps around the World

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1 Big picture

- **Question.** How does the introduction of credit default swaps (CDSs) affect the financing and real decisions of nonfinancial firms, and how do those effects depend on the legal environment in which firms operate?
- **Core idea.** The paper treats the introduction of CDS trading as a shock that changes creditors' bargaining power. But the size of that shock is not universal: it depends on how local bankruptcy law interacts with ISDA credit-event definitions.
- **Main theoretical contribution.** The paper extends Bolton and Oehmke's empty-creditor framework by introducing *legal uncertainty* about whether a firm's action actually triggers CDS payment. Lower uncertainty makes CDS protection more effective and increases firms' debt capacity more strongly.
- **Main empirical contribution.** Using more than 56,000 firms from 51 countries over 2001–2015, the paper shows that CDS introductions raise leverage, but the increase is larger when creditor rights are stronger along trigger-event and liquidation dimensions, when property rights are weaker, and when equity ownership is more concentrated.
- **Headline result.** In the baseline weighted panel regression, CDS introduction raises the leverage ratio by about 0.0123, or roughly 6.8% relative to the sample mean leverage of 0.18.
- **Mechanism.** CDSs strengthen creditors in distress. That extra strength matters most when (i) creditors can more credibly force or influence reorganization, (ii) liquidation payoffs are protected, (iii) contract enforceability is weak so CDSs substitute for poor institutions, and (iv) shareholders are concentrated and otherwise have strong bargaining power.
- **Why this paper matters.** It connects law and finance to the real effects of derivatives. The paper's main message is that CDS contracts are not institution-free: standardized global derivatives still depend on local bankruptcy and enforcement regimes.

2 Data and measurement (key objects)

2.1 Firm-level data and sample

- **Firm sample.** An unbalanced panel of more than 56,000 nonfinancial firms across 51 countries from 2001 to 2015.
- **Equity market data.** Datastream provides monthly stock returns, market capitalization, and local/global market index returns.
- **Accounting data.** WorldScope provides balance-sheet and income-statement variables, including leverage, cash holdings, capital expenditures, R&D, profitability, debt maturity, and closely held shares.
- **CDS data.** Markit provides single-name CDS availability. A firm is treated as a reference entity if it has CDS of any maturity during the year.
- **Treatment definition.** The main treatment is *CDS introduction*: the first year in which CDS are observed on a firm's debt. Because the CDS data start in 2001, introductions are identified beginning in 2002.

- **Sample restrictions.** The paper excludes financial firms (SIC 60–69), firms with nonpositive assets, nonprimary issues, U.S. OTC and Pink Sheet stocks, and firms without usable country or firm identifiers.

2.2 Country-level legal and market variables

- **Creditor rights.** The paper uses the Djankov et al. (2007) creditor-rights variables based on La Porta et al. (1998):
 - *Restrictions on Entry*: creditors can restrict entry into reorganization without their consent;
 - *No Automatic Stay*: no automatic stay or asset freeze protects the borrower from creditors;
 - *Management Does Not Stay*: current management cannot keep administering assets during reorganization;
 - *Secured Creditors First*: secured creditors are paid first out of liquidation proceeds.
- **Property rights.** *Law & Order*, *Corruption*, and *Political Risk* from the ICRG. Higher values mean better property-rights protection and a stronger institutional environment.
- **Credit availability.** Domestic credit to the private sector and total private credit, each scaled by GDP.
- **Financial-market sophistication.** CDS-market depth, stock-market capitalization to GDP, and private-bond-market capitalization to GDP.
- **Ownership concentration.** *Closely Held Shares*, the fraction of equity held by controlling shareholders, from WorldScope.

2.3 Outcome variables

- **Main outcome.** Market leverage, defined as total debt plus preferred stock divided by the market value of total assets.
- **Secondary outcome.** Capital investment, measured as capital expenditures divided by total assets.
- **Interpretation.** The leverage regressions ask whether CDS increase firms’ debt capacity; the investment regressions ask whether the financing change spills over into real asset accumulation.

2.4 Summary patterns in the raw data

- **CDS availability over time.** On average, 1,225 firms per year have available CDSs. Availability rises from 427 firms in 2001 to roughly 1,450 firms per year around 2007–2014.
- **CDS introductions.** There are 1,421 total introductions in the 2002–2015 period, with introductions concentrated before the global financial crisis and falling sharply after 2007.
- **Geography.** The United States and Japan account for more than 62% of the CDS-firm sample, but the paper still has substantial non-U.S. and non-North-American variation.
- **Industries.** CDS availability is especially common in high-PP&E sectors such as utilities, communication, transportation, oil and gas, and chemicals.

- **Selection into treatment.** Firms with CDS introductions are much larger, more levered, have longer debt maturity, and operate in different legal and market environments than untreated firms. This is exactly why the identification strategy matters.

3 Models and empirical specifications (all equations + interpretation)

3.1 Legal uncertainty and the extended Bolton–Oehmke environment

A firm raises debt amount B at time 0 by promising fixed repayment F at time 1. Cash flow at time 1 is either C_1^H with probability θ or C_1^L with probability $(1 - \theta)$, where C_1^L is normalized to zero. After time 1, continuation value is either C_2^H with probability ϕ or C_2^L with probability $(1 - \phi)$.

- **Limited commitment.** Creditors can verify only the low first-period cash flow state without incurring costs, so shareholders may strategically default and force renegotiation.
- **Renegotiation.** In distress, creditors can either liquidate the firm for value S or renegotiate and capture share $q\lambda C_2$ of continuation value, where $\lambda < 1$ measures how much of continuation value survives renegotiation costs and q is creditors' bargaining power.
- **CDS with legal uncertainty.** If creditors hold CDS notional N , their expected protection in the high continuation state is reduced by legal ambiguity about whether a trigger event is recognized:

$$\gamma N = (1 - \varepsilon)N + \varepsilon M, \quad M < N. \quad (1)$$

- **Interpretation.** ε is the probability that the local legal system fails to recognize the relevant trigger event; γ is therefore a reduced-form measure of trigger-event certainty.
- **Economic meaning.** In a creditor-friendly legal environment, γ is higher because creditors can more confidently force a CDS-recognized event.

3.2 Mapping theory to observed country variables

- λ (**contract enforceability / cash-flow verifiability**). Proxied by *Law & Order*, *Corruption*, and *Political Risk*. Weaker property rights imply lower λ .
- q (**creditor bargaining power in private renegotiation**). Inversely related to shareholder control; proxied by *Closely Held Shares*.
- γ (**trigger-event certainty**). Proxied by *Restrictions on Entry*. If creditors must consent before shareholders enter reorganization, trigger-event uncertainty is lower.
- S (**liquidation value**). Proxied by *Secured Creditors First*. Priority in liquidation raises creditors' recovery value.

3.3 Debt capacity without CDS

The firm's debt capacity without CDS is

$$B = \begin{cases} \theta F + (1 - \theta) [\phi q \lambda C_2^H + (1 - \phi) q \lambda C_2^L], & \text{if } F \leq F_C^L, \\ \theta [\phi F + (1 - \phi) q \lambda C_2^L] + (1 - \theta) [\phi q \lambda C_2^H + (1 - \phi) q \lambda C_2^L], & \text{if } F_C^L < F \leq F_C^H, \end{cases} \quad (2)$$

where

$$F_C^L = C_2^L [1 - \lambda(1 - q)], \quad F_C^H = C_2^H [1 - \lambda(1 - q)]. \quad (3)$$

- **Interpretation.** Without CDS, debt capacity is limited by shareholders' incentive to strategically default when honoring the original contract is more expensive than renegotiation.
- **Key force.** Low q or low λ reduces pledgeable income because creditors cannot credibly extract much continuation value in distress.

3.4 Debt capacity with CDS

When creditors hold CDS protection, debt capacity becomes

$$B_{\text{CDS}} = \begin{cases} \theta F + (1 - \theta) [\phi \max(\gamma N, q \lambda C_2^H) + (1 - \phi) N], & \text{if } F \leq \tilde{F}_C^L, \\ \theta [\phi F + (1 - \phi) N] + (1 - \theta) [\phi \max(\gamma N, q \lambda C_2^H) + (1 - \phi) N], & \text{if } \tilde{F}_C^L < F \leq \tilde{F}_C^H, \end{cases} \quad (4)$$

with thresholds

$$\tilde{F}_C^L = C_2^L - \max\{\lambda C_2^L - N, 0\}, \quad \tilde{F}_C^H = C_2^H - \max\{\lambda C_2^H - \gamma N, 0\}. \quad (5)$$

- **Interpretation.** CDSs raise the outside option of creditors in renegotiation.
- **Why legal uncertainty matters.** The high-state outside option is γN , not N . So even if formal CDS notional is large, weak legal recognition of trigger events can mute the debt-capacity gain.
- **Limited-commitment effect.** If $\gamma N > q \lambda C_2^H$ and $N > q \lambda C_2^L$, the CDS contract strengthens creditors enough to relax the firm's borrowing constraint.

3.5 Empty-creditor case

If CDS notional is excessive, so that creditors are overinsured and prefer default to renegotiation, debt capacity becomes

$$B_{\text{CDS}}^{\text{Empty}} = \begin{cases} \theta F + (1 - \theta) [\phi \max(\gamma N, q \lambda C_2^H) + (1 - \phi) S], & \text{if } F \leq \tilde{F}_C^L, \\ \theta [\phi F + (1 - \phi) S] + (1 - \theta) [\phi \max(\gamma N, q \lambda C_2^H) + (1 - \phi) S], & \text{if } \tilde{F}_C^L < F \leq \tilde{F}_C^H, \end{cases} \quad (6)$$

where in this empty-creditor case the thresholds collapse to $\tilde{F}_C^i = C_2^i$ for $i \in \{L, H\}$.

- **Interpretation.** Overinsurance creates a trade-off. CDSs can help in the high state by raising creditors' bargaining power, but in the low state they may induce inefficient liquidation.
- **Why liquidation rules matter.** When secured creditors are paid first, the downside from empty-creditor liquidation is lower, so the net leverage effect of CDS is stronger.

3.6 Proposition 1 and empirical hypothesis

The model's key comparative statics are

$$\frac{\partial \Delta B}{\partial \gamma} \geq 0, \quad \frac{\partial \Delta B}{\partial S} > 0, \quad \frac{\partial \Delta B}{\partial \lambda} < 0, \quad \frac{\partial \Delta B}{\partial q} < 0. \quad (7)$$

- **Interpretation.** CDSs raise debt capacity more when trigger-event certainty is higher, liquidation value is higher, contract enforcement is weaker, and creditors begin with less bargaining power relative to shareholders.
- **Hypothesis 1.** CDS introduction should increase leverage more in countries with less legal uncertainty around triggering events, low liquidation cost, weak contract enforceability, and more concentrated shareholder ownership.

3.7 Propensity score and overlap weighting

Because CDS introduction is endogenous, the paper estimates the probability of treatment as

$$p_{it}(x_t) = \mathbb{P}(Z_{it} = 1 \mid X_{ikt} = x_t), \quad (8)$$

where $Z_{it} = 1$ if firm i receives a CDS introduction in year t and X_{ikt} includes lagged firm and country covariates. The overlap weights are

$$w_{it}(x_t) = \begin{cases} p_{it}(x_t), & \text{if } Z_{it} = 0, \\ 1 - p_{it}(x_t), & \text{if } Z_{it} = 1. \end{cases} \quad (9)$$

- **Interpretation.** Each treated observation is weighted by the probability of being untreated, and each untreated observation is weighted by the probability of being treated.
- **Why this method.** It targets the overlap population, avoids the instability of inverse-probability weights, keeps all observations with positive support, and balances covariates exactly in means.
- **Econometric object.** The paper then estimates the average treatment effect on the treated, ATET, using the reweighted sample.

3.8 Operational weighted panel specification

The leverage and investment tables can be summarized by the weighted panel regression

$$Y_{it} = \alpha + \beta D_{it}^{\text{CDS}} + \delta W_{c(i),t-1} + \eta \left(D_{it}^{\text{CDS}} \times W_{c(i),t-1} \right) + \text{year FE} + \text{industry FE} + u_{it}, \quad (10)$$

where Y_{it} is leverage or investment and $W_{c(i),t-1}$ is one conditioning variable at a time (for example, Restrictions on Entry, Secured Creditors First, Law & Order, Political Risk, or Closely Held Shares).

- **Interpretation of β .** Average effect of CDS introduction in the overlap-weighted sample.
- **Interpretation of η .** How the effect of CDS introduction varies with the legal or ownership environment.
- **Implementation.** All covariates are lagged one year, the regressions include year and Fama–French 48-industry fixed effects, and standard errors are clustered at the firm level.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- The main concern is that CDS introduction is not random. Treated firms differ from untreated firms in size, leverage, debt maturity, ownership concentration, and the legal and financial characteristics of their home countries.
- The paper therefore does *not* interpret a raw comparison of CDS and non-CDS firms as causal.
- Instead, the design tries to compare firms that are similar in observable firm- and country-level characteristics but differ in whether CDS were introduced in that year.

4.2 Why the paper does not rely on standard IVs

- Prior U.S.-focused CDS papers often use lender-side instruments such as banks’ derivative use, capital ratios, or portfolio concentration.
- In a global setting these instruments are harder to defend because they may be correlated with country-level banking regulation, market development, or debt-market depth, all of which can directly affect corporate financing outcomes.
- Restricting the sample to foreign lender-borrower pairs would destroy most of the sample and potentially introduce new selection problems.

4.3 What overlap weighting buys you

- The overlap-weighted sample focuses on firms with substantial probability of both treatment and non-treatment.
- It avoids the sample discard problem of matching and the extreme-weight problem of inverse-probability weighting.
- After weighting, the means of the treated and control groups’ covariates are exactly balanced by construction, which the paper visualizes in Figure 1 and tabulates in Appendix Table E.2.

4.4 Legal interpretation of the identification strategy

- The paper's legal-environment variation is not just generic country fixed heterogeneity. It is tied directly to how local insolvency law maps into the enforceability of CDS contracts.
- Appendix A makes this concrete with the *Abengoa* case in Spain and the *Portugal Telecom/Oi* case in Brazil. Both illustrate that apparently standardized CDS contracts still depend on local bankruptcy provisions when determining whether a credit event has occurred.
- This makes cross-country legal variation economically central rather than merely a robustness split.

4.5 Robustness logic

- The paper also performs a simulation-based sensitivity analysis for omitted variables, following Ichino, Mealli, and Nannicini (2008).
- Additional checks include alternative controls, OLS estimation, excluding U.S. firms, excluding potential national champions, replacing CDS introduction with CDS existence, allowing longer horizons, and testing post-2008 Bang-protocol effects.
- The core leverage results survive these tests in sign and, mostly, in significance.

5 Tables and figures

5.1 Table 1: International CDS introductions and availability

Table 1
International CDS introductions and availability

A. CDS availability by country and year																	Creditor Rights
	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	Average	
Hong Kong	4	5	7	10	14	21	30	37	38	37	39	44	45	45	7	26	4
New Zealand	1	1	1	2	2	2	2	2	2	2	3	3	3	3	2	4	4
United Kingdom	31	48	62	65	66	69	64	59	58	58	60	60	55	58	32	56	4
Australia	9	17	20	23	23	23	22	22	22	22	22	22	23	23	17	21	3
Austria		1	1	2	4	5	5	5	5	6	6	6	6	6	2	4	3
Czech Republic	1	1	1	2	2	2	1	1	1	1	1	1	1	1	1	1	3
Denmark	1	1	1	3	3	3	3	3	3	3	4	4	4	4	4	3	3
Germany	17	20	29	36	37	40	41	43	45	45	44	44	45	45	10	36	3
Ireland	1	1	1	1	1	1	1	1	1	1	2	3	3	3	3	2	3
Korea, Republic Of	4	9	10	11	16	17	18	20	21	23	25	27	27	27	18	3	3
Malaysia	2	2	2	3	6	6	6	7	7	7	7	7	7	7	4	5	3
Netherlands	8	9	13	15	15	16	12	12	12	12	12	12	12	13	1	12	3
Saudi Arabia							1	1	2	2	2	2	2	1	2	3	3
Singapore	1	2	3	3	5	5	7	7	7	7	9	10	10	10	1	6	3
South Africa						1	2	2	4	5	6	6	6	6	4	4	3
Belgium	1	1	4	4	4	4	3	4	4	4	4	4	4	5	4	2	2
Chile						1	2	3	3	5	5	7	7	7	7	5	2
Egypt						1	1	1	1	1	1	1	1	1	1	1	2
India	1	1	1	1	6	15	32	49	50	52	53	52	51	52	43	33	2
Indonesia					2	5	5	5	7	7	8	9	9	9	9	7	2
Italy	7	9	10	14	15	17	18	19	20	20	18	19	22	23	17	2	2
Japan	27	61	121	202	247	267	285	289	288	286	282	278	280	276	255	230	2
Kazakhstan							1	1	1	1	1	1	1	1	1	1	2
Norway	2	2	2	3	5	5	5	7	7	6	6	6	6	6	7	5	2
Romania					1	1	1	1	1	1	1	1	1	1	1	1	2
Russian Federation					2	3	5	5	5	5	5	5	4	4	2	2	2
Spain	5	7	8	11	11	11	13	13	12	10	10	12	12	12	11	2	2

(Continued)

Table 1
Continued

Country	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	Average	Creditor Rights
Sri Lanka								1	1	1	1	1	1	1	1	1	2
Taiwan			6	16	20	24	26	27	27	26	26	28	27	27	23	22	2
Thailand		2	2	2	6	7	10	9	9	10	8	8	8	8	7	7	2
Turkey						2	2	2	2	2	2	1	1	1	1	2	2
Argentina					1	1	1	1	2	3	3	3	3	3	2	1	1
Brazil	9	18	25	37	42	42	39	39	37	37	34	34	32	34	3	31	1
Canada	5	7	6	7	7	7	7	8	8	8	8	7	7	7	7	7	1
Finland	1	2	2	2	2	3	3	3	3	2	2	2	2	2	2	2	1
Greece						1	1	1	1	1	1	1	1	1	1	1	1
Hungary																	
Ireland		1	1	1	2	1	1	1	1	1	1	1	1	1	1	1	1
Philippines			1	2	3	5	6	6	6	6	6	7	7	7	1	5	1
Poland	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Portugal	2	3	3	4	4	4	4	4	4	4	4	4	4	4	5	4	1
Sweden	7	10	10	12	12	14	14	14	14	14	14	13	13	12	13	12	1
Switzerland	4	7	7	9	11	11	12	14	13	13	12	13	13	13	13	11	1
United States	251	347	447	566	615	641	670	661	655	634	626	623	609	599	130	538	1
Colombia							1	1	1	1	1	1	1	1	1	1	0
France	28	36	42	44	45	45	44	44	43	42	43	45	47	48	4	40	0
Mexico				1	1	3	4	7	9	10	12	11	11	11	11	8	0
Bahrain																	
Lebanon						2	1				1	1	1	1	1	1	1
Qatar								1	1	1	1	1	1	1	1	1	1
United Arab Emirates							1	1	1	1	1	1	1	1	1	1	1
Total	427	631	849	1,117	1,271	1,365	1,449	1,483	1,485	1,462	1,455	1,467	1,450	1,445	519	1,225	

(Continued)

Table 1
Continued

B. CDS introductions by country and year																	Creditor Rights
	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	Total		
Hong Kong	1	2	3	4	7	9	7	1		2	5	1			42	4	
New Zealand															2	4	
United Kingdom	18	14	5	2	7	4	4	1	2	2	2				61	4	
Australia	8	3	3								1	1			21	5	
Austria	1	1	1	2	1	3									6	3	
Czech Republic															3	3	
Denmark			2							1					3	3	
Germany	4	9	7	3	3	2	2	2	2						34	3	
Ireland	1									1	1				3	3	
Korea, Republic Of	5	1	1	5	1	1	2	1	2	2	1				23	3	
Malaysia				1	3										5	3	
Netherlands	2	3	2	1	1		1						1		10	3	
Saudi Arabia								1							3	3	
Singapore	1	1	1	2		2				2	1				10	3	
South Africa							2	1							6	3	
Belgium	1	1	3	1			1	1					1		6	2	
Chile	1	1	1	1	1	1									7	2	
Egypt															1	2	
India	1			5	9	17	17	3	2		2		3		57	2	
Indonesia				2	3					2					9	2	
Italy	2	1	5	1	2	1	1	1	1			1	2	1	18	2	
Japan	34	62	81	46	24	20	10	6	1	2		2	1		288	2	
Kazakhstan															1	2	
Norway						1	2								3	2	
Romania															1	2	
Russian Federation						1									5	2	
Spain	2	1	3	2	1	2	1				1				10	2	

(Continued)

Table 1
Continued

B. CDS introductions by country and year																Creditor Rights
	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	Total	
Sri Lanka							1								1	2
Taiwan			6	10	4	4	2								29	2
Thailand		2			4	1	4			1		2			12	2
Turkey															2	2
Argentina															3	1
Brazil				2	4	1	2	3	5	1	1				17	1
Canada	9	7	12	6	3	1	1	1	1				1		42	1
Finland	2	1													4	1
Greece						1									2	1
Hungary															1	1
Ireland	1		1	1	1	2	1								7	1
Philippines		1	1	1	1	2	1								10	1
Poland															3	1
Portugal	1														2	1
Sweden	3			2	2	2	2	9	6	11	19	6	5	11	1	1
Switzerland	105	116	113	74	46	67	15	9	6	11	11	6	5	1	593	1
United States							1								2	0
Colombia															1	0
France	10	6	4	1							2	2	1		28	0
Mexico						2	1	3	2	1					12	0
Lebanon															2	2
Qatar															1	1
United Arab Emirates							1								1	1
Total	218	237	266	187	128	147	80	38	19	31	41	17	11	1	1,421	

Table 1

Continued

C. CDS availability by industry and year

	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	Average
Agriculture	1	1	2	2	4	5	6	7	7	6	7	7	6	8	7	6
Food Products	11	14	20	24	31	33	36	36	36	35	36	38	38	37	18	30
Candy & Soda	4	4	5	9	10	11	12	11	11	8	9	9	9	9	1	8
Brew & Liquor	6	9	13	16	16	18	18	17	17	17	17	13	13	6	14	14
Tobacco Products	9	9	8	9	9	10	10	8	8	8	8	8	8	1	1	8
Recreation	4	7	8	12	13	12	12	12	11	11	11	12	12	10	10	10
Entertainment	5	6	8	12	14	16	19	17	18	18	16	17	16	17	7	14
Printing and Publishing	9	10	13	18	21	24	29	28	18	18	17	17	17	17	9	16
Consumer Goods	15	17	22	24	25	25	25	27	26	26	26	27	27	26	9	23
Apparel	5	6	8	12	12	11	9	10	10	11	9	10	9	1	9	9
Healthcare	3	4	10	13	13	14	14	16	17	17	17	16	16	15	2	12
Medical Equipment	10	7	10	11	12	12	15	15	13	13	13	16	17	17	7	12
Pharmaceutical Products	10	20	26	38	41	45	51	49	43	42	41	40	39	41	12	36
Rubber and Plastic Products	19	30	40	54	65	64	64	68	69	70	67	67	68	68	25	56
Chemicals	1	1	3	3	3	3	4	5	5	5	5	4	4	4	3	4
Textiles	7	1	2	4	4	4	5	6	6	6	6	6	5	5	4	5
Construction Materials	7	15	19	27	31	30	34	37	37	37	36	36	35	35	13	29
Construction	3	11	17	27	33	42	56	62	64	64	64	68	66	67	31	45
Steel Works Etc	10	13	18	29	35	35	38	44	43	44	45	46	46	46	25	34
Fabricated Products	10	25	35	45	50	55	56	56	56	55	56	56	57	33	1	47
Machinery	3	6	10	12	17	18	21	22	22	21	21	18	18	18	10	16
Electrical Equipment	23	34	39	47	56	60	63	64	63	60	61	63	63	60	22	52
Automobiles and Trucks																

(Continued)

Table 1

Continued

Aircraft		8	10	11	11	11	10	10	11	11	11	11	11	1	10
Shipbuilding, Railroad Equipment		1	1	1	2	2	2	2	3	4	4	3	4	2	3
Defense		1	1	1	2	2	2	2	2	2	2	2	2	2	2
Precision Metals		1	1	1	3	3	3	3	3	3	3	3	3	3	3
Non-Metallic and Industrial Metal Mining		3	8	13	15	16	15	17	18	18	19	19	18	19	6
Coal		27	45	55	63	66	76	81	81	80	77	82	80	83	9
Petroleum and Natural Gas		43	51	87	117	137	148	150	153	150	146	147	141	143	139
Utilities		46	55	74	79	92	97	101	104	107	103	102	106	105	102
Communication		1	3	3	4	6	6	7	7	7	8	8	9	7	8
Personal Services		13	21	34	52	63	68	73	73	72	75	75	80	77	78
Business Services		9	9	14	19	23	27	28	27	28	27	26	27	27	27
Computers		15	22	38	57	65	71	69	71	71	68	67	62	61	60
Electronic Equipment		4	8	10	12	13	13	14	14	14	14	13	13	13	8
Measuring and Control Equipment		11	12	14	21	24	26	32	32	34	34	31	33	33	11
Business Supplies		4	5	10	12	12	12	13	13	13	14	14	14	14	13
Transportation		21	35	52	75	79	85	87	87	89	88	88	91	94	93
Wholesale		8	14	19	22	31	36	37	40	38	39	38	36	36	21
Retail		37	44	53	69	74	81	92	94	91	90	87	85	85	34
Restaurants, Hotels, Motels		10	15	20	26	26	30	29	31	31	31	32	33	33	13
Other Industries		3	4	6	6	6	7	8	9	9	9	7	7	6	7
Total		427	631	849	1,117	1,271	1,365	1,449	1,483	1,485	1,462	1,455	1,467	1,450	1,445

The table shows the number of CDS reference entities by year across countries (panel A) and industries (panel B) and the number of CDS introductions by year across countries (panel B). Countries in panels A and B are sorted by the creditor rights index as reported in the last column of panel A. The sample consists of an unbalanced panel of more than 56,000 nonfinancial firms across 51 countries over the period 2001–2015. Market data are from Datastream; accounting data are from WorldScope; and CDS data are from Markit.

Table 2

Propensity for CDS introduction

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
	Creditor Rights				Property Rights				Private Credit Availability					
	Restriction on Entry	No Automatic Stay	No Management Does Not Stay	Secured Creditors First	Law and Order	Corruption	Political Risk	Domestic Credit in Private Sector	Private Credit	CDS	Stock Market	Private Bond Market	Closely Held Shares	
Country/Firm Variable	-0.129*** (0.042)	-0.129*** (0.048)	-0.310*** (0.051)	-0.129*** (0.049)	0.241*** (0.048)	-0.009 (0.041)	-0.018 (0.049)	0.329*** (0.062)	-0.068 (0.054)	0.450*** (0.145)	0.118*** (0.045)	0.276*** (0.049)	-0.259*** (0.041)	
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Clustered Standard Errors	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	
Observations	80,822	80,822	80,822	80,822	80,822	80,822	80,822	80,822	80,822	80,822	80,822	80,822	80,822	
Pseudo R-Squared	0.397	0.396	0.400	0.397	0.399	0.396	0.396	0.400	0.396	0.401	0.396	0.400	0.400	

The table shows the results of logit regressions in which the CDS introduction dummy (i.e., only the first year of CDS trading for each firm) is the dependent variable. Firm characteristics and country characteristics serve as explanatory variables, and all are lagged by one year. (Standardized) country variables are grouped into four categories: (1) creditor rights, (2) property rights, (3) private credit availability, and (4) financial market sophistication. The last column uses the (standardized) firm-level variable *Closely Held Shares* instead of a country variable. For creditor rights, we use an aggregate index of four (Creditor Rights) as well as its four subindices, namely, restrictions on a firm entering organization without creditors' consent (*Restrictions on Entry*); no automatic stay or asset freeze (*No Automatic Stay*); restriction on management's administration of a firm's assets pending resolution of the reorganization (*Management Does Not Stay*); and payment of secured creditors first out of any liquidation proceeds (*Secured Creditors First*). For property rights, we consider the following three measures: *Law and Order*, *Corruption*, and *Political Risk*, where higher index values indicate stronger protection for private properties. Private credit availability is measured by domestic credit by financial corporations to the private sector scaled by GDP (*Domestic Credit in Private Sector*) and by total credit to the private nonfinancial sector scaled by GDP (*Private Credit*). Financial market sophistication is measured by the ratio of market capitalization of CDS firms to that of all firms in a country (*CDS Market*), stock market capitalization scaled by GDP (*Stock Market*), and market capitalization of the private bond market scaled by GDP (*Private Bond Market*). For equity ownership concentration, we use the percentage of closely held shares (*Closely Held Shares*). All regressions include the following firm characteristics as controls: the natural logarithm of total assets (in USD), *Tobin's q*, the market-to-book equity ratio, return on assets (3-year average), gross profit margin (3-year average), a dividend dummy, cash flow to sales, free cash flow to total assets, the natural logarithm of cash and short-term investments, capital expenditures to total assets, R&D to assets, net PPE to size, market leverage, industry median market leverage, convertible debt to size, debt maturity, earnings volatility in local currency and in USD, volatility of return on assets, net foreign exchange exposure, firm age, and tax rate. Regressions also include year and Fama-French 48-industry fixed effects. Standard errors are clustered at the firm level and are reported in parentheses. The sample consists of an unbalanced panel of more than 56,000 nonfinancial firms across 51 countries over the period 2001–2015. Appendix D defines all variables. * $p < .1$; ** $p < .05$; *** $p < .01$.

- This is the paper's descriptive map of where CDSs exist and when they are introduced.
- The key takeaways are: CDSs are concentrated in developed markets, especially the United States and Japan; introductions boom before the financial crisis and slow sharply afterward; and there is still meaningful non-U.S. variation, which is crucial for the law-and-finance question.
- Panel C also shows that CDSs are more common in capital-intensive industries, which makes sense because those sectors rely more heavily on external debt financing.

5.2 Table 2: Propensity for CDS introduction

Read:

- This table documents the selection problem the paper must solve.
- CDS introduction is *less* likely in stronger creditor-rights environments on most dimensions: the coefficients on the aggregate creditor-rights index, Restrictions on Entry, No Automatic Stay, and Management Does Not Stay are all negative and significant.
- The exception is Secured Creditors First, which is positive and significant, suggesting that creditor priority in liquidation makes CDS introduction more attractive.
- Domestic credit availability and financial-market sophistication strongly predict CDS introduction, while closely held shares predict less treatment. So treated firms differ in both firm characteristics and institutional environments.

5.3 Table 3: Firm- and country-level characteristics without overlap weights

Read:

- This table shows the raw imbalance between treated and control observations.

- Treated firms are much larger, more levered, have longer debt maturity, and operate in countries with systematically different legal and market characteristics.
- The message is that untreated firms are a bad counterfactual in the raw data. Any causal claim needs reweighting or some other adjustment.

	Treated (N = 784)		Control (N = 80,038)		% Bias	t-test		Kolmogorov-Smirnov test	
	Mean	Std. Err.	Mean	Std. Err.		t	p-value	D	p-value
Firm Characteristics									
Cash Flow/Sales	0.149	0.004	0.112	0.001	285	6.7	0.000	0.1399	0.000
Coverable Debt/Size	0.011	0.001	0.002	0.000	535	22.2	0.000	0.2061	0.000
Debt Maturity	0.736	0.008	0.528	0.001	795	18.8	0.000	0.3084	0.000
Dividend	0.835	0.013	0.758	0.002	199	5.0	0.000	0.0771	0.000
Free Cash Flow/Total Assets	0.025	0.002	0.026	0.000	-2%	-0.4	0.687	0.0647	0.003
Gross Profit Margin (3y)	0.306	0.006	0.271	0.001	209	5.4	0.000	0.1027	0.000
Leverage Market Value	0.342	0.005	0.190	0.001	335	9.0	0.000	0.2607	0.000
Leverage Market Value (Industry Median)	0.167	0.003	0.152	0.000	209	5.7	0.000	0.1056	0.000
Age (log)	2.791	0.027	2.352	0.003	325	9.0	0.000	0.2779	0.000
ROA Volatility (log)	-3.900	0.031	-3.541	0.004	-38%	-10.1	0.000	0.1498	0.000
Total Assets in USD (log)	21.773	0.023	19.326	0.006	189%	40.3	0.000	0.6955	0.000
Market/Book	2.479	0.069	1.943	0.007	28%	8.0	0.000	0.2350	0.000
Net FX Exposure	0.119	0.008	0.117	0.001	1%	0.2	0.859	0.0751	0.000
PPE (Net)/Size	0.395	0.010	0.391	0.001	2%	0.4	0.691	0.0615	0.006
Returns On Assets (3y)	0.062	0.002	0.061	0.000	3%	0.7	0.566	0.0520	0.230
Tax Rate	0.344	0.005	0.304	0.001	27%	7.0	0.000	0.1795	0.000
Tobin's Q	1.321	0.034	1.181	0.004	14%	3.8	0.000	0.2125	0.000
Return Volatility in LC (log)	-1.146	0.015	-0.969	0.002	-43%	-10.6	0.000	0.1812	0.000
Return Volatility in USD (log)	-1.108	0.015	-0.915	0.002	-46%	-11.6	0.000	0.1989	0.000
Capital Expenditure/Total Assets	0.060	0.002	0.051	0.000	18%	4.8	0.000	0.1565	0.000
R&D/Total Assets	0.014	0.001	0.010	0.000	15%	4.5	0.000	0.1173	0.000
Cash and Short-term Investments/Total Assets (log)	-2.721	0.046	-2.383	0.005	-26%	-7.1	0.000	0.1369	0.000
Closely Held Shares	-0.628	0.035	0.040	0.004	-67%	-18.0	0.000	0.3007	0.000
Country Characteristics									
Restrictions on Entry	-0.309	0.030	0.091	0.004	-23%	-8.0	0.000	0.1304	0.000
No Automatic Stay	-0.300	0.029	0.091	0.004	1%	-10.4	0.000	0.1831	0.000
Management Does Not Stay	-0.141	0.036	0.065	0.004	18%	-5.8	0.000	0.1027	0.000
Secured Creditors First	0.105	0.030	0.006	0.004	10%	2.8	0.005	0.0304	0.468
Law & Order	0.280	0.034	-0.054	0.004	49%	8.1	0.000	0.1330	0.000
Corruption	0.060	0.028	-0.002	0.003	37%	2.6	0.009	0.1531	0.000
Political Risk	0.386	0.028	0.017	0.003	1%	10.7	0.000	0.2339	0.000
Domestic Credit to Private Sector	0.344	0.032	0.044	0.003	20%	8.6	0.000	0.2517	0.000
Private Credit	0.022	0.029	-0.026	0.004	-6%	1.4	0.174	0.1378	0.000
CDS Market	0.225	0.030	0.062	0.003	29%	5.2	0.000	0.2541	0.000
Stock Market	0.030	0.023	0.088	0.004	-10%	-1.4	0.175	0.1369	0.000
Private Bond Market	0.339	0.037	-0.088	0.003	13%	12.4	0.000	0.2479	0.000

The table compares firm- and country-level characteristics between firm-years with CDS introductions (treated) and without CDS introductions (controls) in the prior year. It reports the mean and standard errors for treated and control firms; the percentage bias according to Rosenbaum and Rubin (1987); and test statistics and *p*-values of *t*-tests and Kolmogorov-Smirnov tests. The sample consists of an unbalanced panel of more than 56,000 nonfinancial firms across 51 countries over the period 2001–2015. Market data are from Datastream; accounting data are from WorldScope; and CDS data are from Markit. The sample is limited to firm-year observations for which all reported firm- and country-level variables are jointly available. Given the differences in characteristics between treated and control firms, we use overlap weights (Li, Morgan, and Zaslavsky 2010) for our main analysis that balance the covariates between these

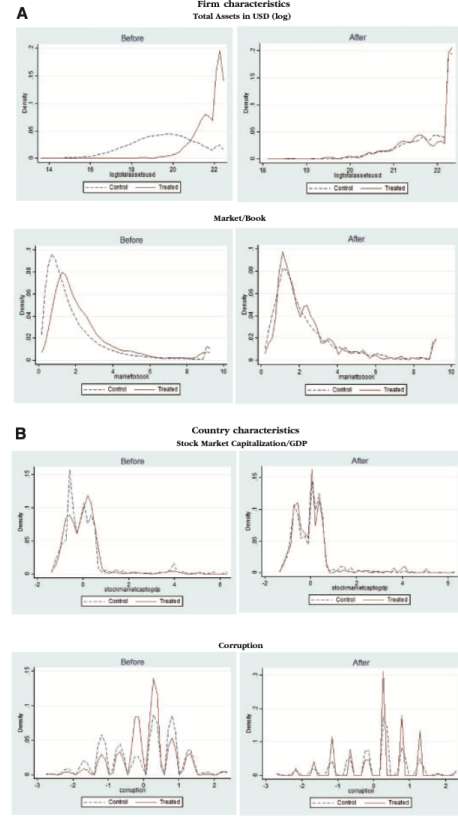


Figure 1
Covariate balancing of sample firms

This figure shows the covariate balancing of sample firms a year prior to CDS introduction by plotting the distributions for treated firms (i.e., firms in the year of CDS introduction) and control firms (i.e., firms without CDS introductions in that year) before and after imposing overlap weights. Panel A shows results for selected firm characteristics; panel B shows results for selected country characteristics. Variables are selected based on the largest EADD in Table E.2 in Appendix E. The sample consists of an unbalanced panel of more than 56,000 nonfinancial firms across 51 countries over the period 2001–2015. Market data are from Datastream; accounting data are from WorldScope; and CDS data are from Markit. Appendix D defines all variables.

5.4 Figure 1: Covariate balancing of sample firms

Read:

- Figure 1 is the visual payoff of the overlap-weighting design.
- Before weighting, treated and control densities differ sharply for firm size, market-to-book, stock-market depth, and corruption. After weighting, the densities line up closely.
- This figure is important because it shows that the paper is not simply running weighted regressions mechanically; it is actually creating a credible overlap sample.

5.5 Table 4: Effects of CDSs on leverage

Read:

- Column 1 gives the baseline weighted treatment effect: CDS introduction raises leverage by 0.0123.
- The interaction terms are the heart of the paper. The CDS effect is stronger when Restrictions on Entry is high, when Secured Creditors First is high, when Law & Order or Political Risk is low, and when Closely Held Shares is high.
- Aggregate creditor rights are not enough; what matters are the specific dimensions tied to trigger-event certainty and liquidation payoffs.
- The property-rights result is especially interesting: CDSs have larger effects where contract enforceability is weaker, meaning they partly substitute for poor institutions.

Table 4
Effects of CDSs on leverage

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Creditor Rights					Property Rights				
	Baseline	Creditor Rights	Restriction on Entry	No Automatic Stay	Management Does Not Stay	Secured Creditors First	Law & Order	Corruption	Political Risk	Closely Held Shares
CDS Introduction × Conditioning Variable		0.0028	0.0128**	-0.0063	-0.0069	0.0149**	-0.0190***	-0.0080	-0.0304***	0.0134**
		(0.0055)	(0.0065)	(0.0063)	(0.0066)	(0.0057)	(0.0072)	(0.0074)	(0.0077)	
Conditioning Variable		0.0003	-0.0120***	0.0018	0.0125***	-0.0095***	-0.0002	-0.0112***	0.0065	0.0126***
		(0.0037)	(0.0040)	(0.0030)	(0.0040)	(0.0032)	(0.0036)	(0.0041)	(0.0047)	(0.0038)
CDS Introduction	0.0123**	0.0129**	0.0144**	0.0106*	0.0115**	0.0111**	0.0156***	0.0128**	0.0225***	0.0198***
	(0.0056)	(0.0057)	(0.0057)	(0.0056)	(0.0057)	(0.0057)	(0.0056)	(0.0055)	(0.0060)	
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered Standard Errors	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm
Observations	80,501	80,501	80,501	80,501	80,501	80,501	80,501	80,501	80,501	80,501
Adj. R-Squared	0.187	0.187	0.189	0.188	0.191	0.189	0.193	0.193	0.195	0.202

The table shows the average treatment effect of CDS introduction on the market leverage (defined as the sum of total debt and preferred stock divided by market value of total assets) of the treated firms (ATET) using overlap weights (Li, Morgan, and Zaidavsky 2018). The treatment is the introduction of CDS (i.e., only the first year of CDS trading for each firm). The regressions further include CDS interaction effects with lagged (standardized) conditioning variables, as well as the lagged conditioning variables themselves. The use of overlap weights ensures that covariates are perfectly balanced between treated firms and control firms in the year before treatment (see table E.2 in Appendix A). Conditioning variables are grouped into three categories: (1) creditor rights, (2) property rights, and (3) equity ownership concentration. For creditor rights, we use an aggregate index (*Creditor Rights*) as well as its four subdimensions, namely, restrictions on a firm entering reorganization without creditors' consent (*Restrictions on Entry*); no automatic stay or asset freeze (*No Automatic Stay*); restriction on management's administration of a firm's assets pending resolution of the reorganization (*Management Does Not Stay*); and payment of secured creditors first out of any liquidation proceeds (*Secured Creditors First*). For property rights, we consider the following three measures: *Law & Order*, *Corruption*, and *Political Risk*, where higher index values indicate stronger protection for private properties. For equity ownership concentration, we use the percentage of closely held shares (*Closely Held Shares*). All regressions include year and Fama-French 48-industry fixed effects. Standard errors are clustered at the firm level and are reported in parentheses. The sample consists of an unbalanced panel of more than 56,000 nonfinancial firms across 51 countries over the period 2001–2015. Market data are from Datastream; accounting data are from WorldScope; and CDS data are from Markit. Appendix D defines all variables. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 5
Sensitivity analysis for omitted variables

Sensitivity analysis for omitted variables											
		(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	
		Creditor Rights					Property Rights				
		Creditor Rights	Restriction on Entry	No Automatic Stay	Management Does Not Stay	Secured Creditors First	Law & Order	Corruption	Political Risk	Closely Held Shares	
Calibrated Confounders	Minimum	0.0025	0.0124*	-0.0071	-0.0071	0.0147**	-0.0191***	-0.0086	-0.0304***	0.0132**	
	Maximum	0.0031	0.0131**	-0.0057	-0.0066	0.0152***	-0.0182***	-0.0071	-0.0294***	0.0141**	
Killer Confounders	Minimum	0.0017	0.0100	-0.0085	-0.0087	0.0118*	-0.0211***	-0.0084	-0.0242***	0.0130**	
	Maximum	0.0042	0.0155**	-0.0034	-0.0053	0.0152***	-0.0158***	-0.0056	-0.0255***	0.0161**	

The table reports tests of the sensitivity of the regression results in table 4 to the effect of unobserved omitted variables in the estimation of the balancing weights, using a simulation approach proposed by Ichim, Mealli, and Nannicini (2008). The regression specifications (2) to (10) are the same as in table 4. The key treatment indicator is the CDS introduction dummy. Its interaction effects with (standardized) conditioning characteristics are also jointly estimated. Conditioning variables are grouped into three categories: (1) creditor rights, (2) property rights, and (3) equity ownership concentration. For creditor rights, we use an aggregate index (*Creditor Rights*) as well as its four subdimensions, namely, restrictions on a firm entering reorganization without creditors' consent (*Restrictions on Entry*); no automatic stay or asset freeze (*No Automatic Stay*); restriction on management's administration of a firm's assets pending resolution of the reorganization (*Management Does Not Stay*); and payment of secured creditors first out of any liquidation proceeds (*Secured Creditors First*). For property rights, we consider the following three measures: *Law & Order*, *Corruption*, and *Political Risk*, where higher index values indicate stronger protection for private properties. For equity ownership concentration, we use the percentage of closely held shares (*Closely Held Shares*). The sensitivity analysis simulates alternatively calibrated confounders or killer confounders in the estimation of the balancing weights that are subsequently used to assess the average treatment effect on the treated (ATET) using overlap weights (Li, Morgan, and Zaidavsky 2018). Calibrated confounders are specified to have an empirical distribution similar to the existing observable covariates in the logit regression that yields the inputs to the balancing weights. We use binary transformations of continuous covariates, that is, indicator variables of whether an observation is above or below the median of that variable. Separately, we also use killer confounders to assess whether more extreme unobserved omitted covariates exist that eliminate the treatment effect. We simulate 100 iterations of calibrated and killer confounders and reestimate the effect of CDS introduction and their interactions with conditioning variables for each iteration. Further details on the methodology are provided in Appendix B. For each outcome regression specification, the table reports the minimum and maximum coefficients for the interaction variables across calibrated and killer confounders. Standard errors are clustered at the firm level. The sample consists of an unbalanced panel of more than 56,000 nonfinancial firms across 51 countries over the period 2001–2015. All variables are defined in Appendix D. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.6 Table 5: Sensitivity analysis for omitted variables

Read:

- The paper simulates both calibrated confounders and more extreme “killer confounders.”
- The signs of the main interaction effects are preserved, and most significance levels remain intact.
- The only coefficients that become somewhat more fragile under extreme confounders are Restrictions on Entry and Secured Creditors First, but even there the point estimates do not flip sign. So the qualitative conclusions are robust.

5.7 Selected appendix exhibits: investment, ex-U.S. sample, and Bang protocols

Read:

- **Appendix Table E.3.** The unconditional investment effect of CDS introduction is small, but the interaction with Restrictions on Entry is positive and highly significant. This suggests that the extra borrowing capacity translates into more capital investment when creditors have stronger control over reorganization.
- **Appendix Table E.6.**

Table E.3

Effects of CDS on capital investment

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Creditor Rights					Property Rights				
	Baseline	Creditor Rights	Restriction on Entry	No Automatic Stay	Management Does Not Stay	Secured Creditors First	Law & Order	Corruption	Political Risk	Closely Held Shares
CDS Introduction × Conditioning Variable	0.0031*	0.0053***	0.0001	-0.0006	0.0017	-0.0008	-0.0003	-0.0046	0.0023	
Conditioning Variable	-0.0007	-0.0010	-0.0015	-0.0009	0.0007	-0.001**	-0.0016	-0.0041**	0.0017	
CDS Introduction	0.0010	0.0012	0.0011	0.0011	0.0009	0.0011	0.0013	0.0012	0.0028	0.0001
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered Standard Errors	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm
Observations	80,822	80,822	80,822	80,822	80,822	80,822	80,822	80,822	80,822	80,822
Adj. R-Squared	0.263	0.265	0.266	0.264	0.264	0.265	0.268	0.264	0.274	0.267

The table shows the average treatment effect of CDS introductions on the capital investment (defined as capital expenditures divided by total assets) of the treated firms (ATET) using overlap weights (Li, Morgan, and Zaslavsky 2018). The treatment is the introduction of CDS (i.e., only the first year of CDS trading for each firm). The regressions further include CDS interaction effects with lagged (standardized) conditioning variables, as well as the lagged conditioning variables themselves. The use of overlap weights ensures that covariates are perfectly balanced between treated firms and control firms in the year before treatment (see Table E.2 in Appendix E). Conditioning variables are grouped into three categories: (1) creditor rights, (2) property rights, and (3) equity ownership concentration. For creditor rights, we use an aggregate index (*Creditor Rights*) as well as its four subindices, namely, restriction on a firm entering reorganization without creditors' consent (*Restrictions on Entry*); no automatic stay or asset freeze (*No Automatic Stay*); restriction on management's administration of a firm's assets pending resolution of the reorganization (*Management Does Not Stay*); and payment of secured creditors first out of any liquidation proceeds (*Secured Creditors First*). For property rights, we consider the following three measures: *Law & Order*, *Corruption*, and *Political Risk*, where higher index values indicate stronger protection for private property. For equity ownership concentration, we use the percentage of closely held shares (*Closely Held Shares*). All regressions include year and Fama-French 48-industry fixed effects. Standard errors are clustered at the firm level and are reported in parentheses. The sample consists of an unbalanced panel of more than 56,000 nonfinancial firms across 51 countries over the period 2001–2015. Market data are from Datastream; accounting data are from WorldScope; and CDS data are from Markit. All variables are defined in Appendix D. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table E.6

Results excluding U.S. firms

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Creditor Rights					Property Rights				
	Baseline	Creditor Rights	Restriction on Entry	No Automatic Stay	Management Does Not Stay	Secured Creditors First	Law & Order	Corruption	Political Risk	Closely Held Shares
CDS Introduction × Conditioning Variable	0.0005	0.0190***	-0.0021	-0.0083	0.0149**	-0.0185***	-0.0080	-0.0246***	0.0218***	
Conditioning Variable	-0.0049*	-0.0070	-0.0070	-0.0086	-0.0065	-0.0064	-0.0060	-0.0077	-0.0075	
CDS Introduction	0.0115	0.0099	0.0099	0.0097	0.0097	0.0099	0.0097	0.0090	0.0091	0.0041
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered Standard Errors	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm
Observations	65,847	65,847	65,847	65,847	65,847	65,847	65,847	65,847	65,847	65,847
Adj. R-Squared	0.207	0.212	0.216	0.209	0.208	0.210	0.212	0.213	0.217	0.214

The table shows the average treatment effect of CDS introductions for non-US firms on the market leverage (defined as the sum of total debt and preferred stock divided by the market value of total assets) of the treated firms (ATET) using overlap weights (Li, Morgan, and Zaslavsky 2018). The treatment is the introduction of CDS (i.e., only the first year of CDS trading for each firm). The regressions further include CDS interaction effects with lagged (standardized) conditioning variables, as well as the lagged conditioning variables themselves. The use of overlap weights ensures that covariates are perfectly balanced between treated firms and control firms in the year before treatment (see Table E.2 in Appendix E). Conditioning variables are grouped into three categories: (1) creditor rights, (2) property rights, and (3) equity ownership concentration. For creditor rights, we use an aggregate index (*Creditor Rights*) as well as its four subindices, namely, restriction on a firm entering reorganization without creditors' consent (*Restrictions on Entry*); no automatic stay or asset freeze (*No Automatic Stay*); restriction on management's administration of a firm's assets pending resolution of the reorganization (*Management Does Not Stay*); and payment of secured creditors first out of any liquidation proceeds (*Secured Creditors First*). For property rights, we consider the following three measures: *Law & Order*, *Corruption*, and *Political Risk*, where higher index values indicate stronger protection for private property. For equity ownership concentration, we use the percentage of closely held shares (*Closely Held Shares*). All regressions include year and Fama-French 48-industry fixed effects. Standard errors are clustered at the firm level and are reported in parentheses. The sample consists of an unbalanced panel of nonfinancial firms across 50 countries over the period 2001–2015. Market data are from Datastream; accounting data are from WorldScope; and CDS data are from Markit. All variables are defined in Appendix D. * $p < .1$; ** $p < .05$; *** $p < .01$.

Excluding U.S. firms leaves the main leverage pattern intact. The results are therefore not just a relabeling of U.S. evidence.

- **Appendix Table E.7.** After the 2009 Big/Small Bang protocol changes, the importance of

Table E.7

Role of the bang protocols on the effect on leverage of a CDS introduction

A. Full sample

	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	Creditor Rights	Restriction on Entry	No Automatic Stay	Management Does Not Stay	Secured Creditors First
CDS Introduction × Conditioning Variable × Post-Bang Dummy	0.0148	-0.0272	0.0235	0.0287*	0.0136	0.0189
CDS Introduction × Conditioning Variable	0.0173	0.0224	0.0212	0.0221	0.0041	0.0152
CDS Introduction × Post-Bang Dummy	0.0024	0.0196***	-0.0059	-0.0100*	0.0096	0.0098
CDS Introduction	0.0149	0.0155	0.0146	0.0197	0.0152	0.0073
Post-Bang Dummy × Conditioning Variable	0.0022	0.0221	0.0217	0.0223	0.0042	0.0077
Conditioning Variable	0.0033	0.0144**	0.0098*	-0.0117**	0.0024	-0.0076
Year Fixed Effects	-0.0081	-0.0007	-0.0044	-0.0005	-0.0044	-0.0007
Industry Fixed Effects	0.0060	0.0060	0.0059	0.0060	0.0060	0.0060
Clustered Standard Errors	80,501	80,501	80,501	80,501	80,501	80,501
Observations	0.156	0.157	0.159	0.158	0.160	0.158

(Continued)

Table E.7

Continued

B. Sample excluding North American firms

	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	Creditor Rights	Restriction on Entry	No Automatic Stay	Management Does Not Stay	Secured Creditors First
CDS Introduction × Conditioning Variable × Post-Bang Dummy	0.0043	-0.0545**	0.0149	0.0440	0.0189	0.0189
CDS Introduction × Conditioning Variable	0.0191	0.0228	0.0245	0.0245	0.0152	0.0152
CDS Introduction × Post-Bang Dummy	0.0078	0.0076	0.0076	0.0076	0.0098	0.0098
Post-Bang Dummy × Conditioning Variable	0.0094	0.0151**	0.0158	0.0121*	-0.0189**	0.0027
Conditioning Variable	0.0042	-0.0244***	-0.0068	-0.0068	0.0073	-0.0077**
Year Fixed Effects	-0.0083	-0.0105	-0.0095	-0.0090	-0.0002	-0.0076
Industry Fixed Effects	0.0073	0.0073	0.0073	0.0073	0.0073	0.0073
Clustered Standard Errors	64,146	64,146	64,146	64,146	64,146	64,146
Observations	0.156	0.170	0.169	0.169	0.168	0.168

The table shows the average treatment effect of CDS introductions on the market leverage (defined as the sum of total debt and preferred stock divided by market value of total assets) of the treated (ATET) using overlap weights (Li, Morgan, and Zaslavsky 2018) for the full sample (panel A) and for sample firms excluding firms from North America (panel B), respectively. The treatment is the introduction of CDS (i.e., only the first year of CDS trading for each firm). The post-bang dummy has a value equal to 1 for years after 2008 (and zero otherwise). The regressions further include interactions between CDS introduction, Post-Bang Dummy, and lagged (standardized) conditioning variables, as well as the lagged conditioning variables themselves. The use of overlap weights ensures that covariates are perfectly balanced between treated firms and control firms in the year before treatment (see Table E.2 in Appendix E). Conditioning variables are an aggregate index (*Creditor Rights*) as well as its four subindices, namely, restriction on a firm entering reorganization without creditors' consent (*Restrictions on Entry*); no automatic stay or asset freeze (*No Automatic Stay*); restriction on management's administration of a firm's assets pending resolution of the reorganization (*Management Does Not Stay*); and payment of secured creditors first out of any liquidation proceeds (*Secured Creditors First*). All regressions include year and Fama-French 48-industry fixed effects. Standard errors are clustered at the firm level and are reported in parentheses. The sample consists of an unbalanced panel of nonfinancial firms across 51 countries over the period 2001–2015. Market data are from Datastream; accounting data are from WorldScope; and CDS data are from Markit. All variables are defined in Appendix D. * $p < .1$; ** $p < .05$; *** $p < .01$.

Restrictions on Entry declines, especially outside North America. This is consistent with the interpretation that the Bang protocols reduced legal uncertainty around CDS trigger events.

6 Short summary

- **Conclusion.** CDS introduction raises firms' leverage, but the size of the effect depends strongly on the legal environment.
- **Intuition.** CDSs strengthen creditors in distress; that matters more when bankruptcy law makes the trigger event more credible, when liquidation rights are protected, when property rights are weak, and when shareholders are powerful.

- **Empirical lesson.** In a cross-country setting, legal detail is not a side issue. It is the mechanism through which a global derivative contract changes real corporate behavior.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** CDS introduction increases leverage for the underlying nonfinancial firm.
- **Magnitude.** The baseline effect is economically meaningful: about a 6.8% increase in leverage relative to the sample mean.
- **Where the effect is strongest.** The leverage response is largest when creditors can restrict entry into reorganization, when secured creditors are paid first, when property rights are weak, and when ownership is concentrated.
- **Secondary real effect.** There is also evidence that CDSs raise capital investment in settings where creditor control over reorganization is strong.
- **Robustness.** The paper's conclusions survive overlap balancing, omitted-variable sensitivity checks, extra controls, OLS, an ex-U.S. sample, and post-Bang comparisons.

7.2 Economic intuition

- CDSs sever part of the usual link between holding debt and wanting efficient continuation of the firm. That is the empty-creditor logic.
- But this force is not uniform across countries because CDS payoffs depend on whether local law makes the relevant credit event recognizable and enforceable.
- In weak institutional environments, CDSs can substitute for poor enforcement by giving creditors a stronger outside option. In stronger trigger-event environments, they also let creditors extract more value in renegotiation, which supports more debt ex ante.

7.3 Takeaway

This paper's key contribution is to show that the real effects of CDSs are mediated by law. A standardized derivative contract does not produce standardized economic effects when bankruptcy, liquidation, and enforcement differ across countries. In that sense, the paper is both a CDS paper and a law-and-finance paper: it shows how global derivatives interact with local institutions to shape leverage and, to a lesser extent, investment.